The remainder of Probst and Holladay's recommendations are essentially identical to those that we made in our article.

DOUGLAS D. KOCH, MD
Houston, Texas

Sensitivity and Specificity

Dear Editor:
The definitions of sensitivity and specificity provided by Choplin and Lundy[1] lead to nonsense. Consider this table:

	Disease		
Diagnosis	*Normal*	*Suspect*	*Glaucoma*
Normal	a	b	c
Suspect	d	e	f
Glaucoma	g	h	i

"Disease" indicates the true condition of the patient. Here, a through i are probabilities that sum to 1. Choplin and Lundy define false positive as (d+g+h), false negative as (b+c+f), sensitivity as (a+d+e+g+h+i), and specificity as (a+b+c+e+f+i). Let us assume that a = 0.8, b = c = 0.1, and all other cells = 0. According to Choplin and Lundy, the sensitivity would be 0.8, even though no subject with disease is ever detected. Now assume that a = 0, b = c = 0.4, d = g = 0.1, and all other cells = 0. According to Choplin and Lundy, the specificity would be 0.8, even though no normal subject is ever diagnosed correctly.

More reasonable definitions would be sensitivity = (e+h+i)/(b+c+e+f+h+i) and specificity = a/(a+d+g).

TERRY A. COX, MD, PHD
Bethesda, Maryland

Author reply

Dear Editor:
We appreciate Dr. Cox's interest in our work and the description of a tabulation for our data. It is particularly interesting to note that we had used this exact tabulation format in the original submission of our article! Perhaps we were not adept in describing the data in a clear and forthright manner, but the reviewers suggested a revision with alternative displays. The overall feeling was that the format using the tabulation suggested by Dr. Cox was too confusing.

We do not understand Dr. Cox's statement about no normal subject being diagnosed correctly using his model. If box a equals 0.8, then 80% of the time a normal patient was called normal (i.e., the specificity would be 80%), and 80% of the normals were therefore diagnosed correctly. Assuming boxes b, c, e, f, h, and i were indeed 0, there were no patients who were not normal and no statement could be made regarding sensitivity. In our population of normal subjects, those suspected to have glaucoma, and patients with glaucoma, this sort of distribution would not have occurred.

We do understand that our definitions of sensitivity and specificity are unconventional, and thank Dr. Cox for pointing this out. The false-positive rate should be how many times a normal patient is classified under "disease" (with specificity being 1 minus the false-positive rate) and the false-negative rate should be how often a patient with disease is called normal (with sensitivity being 1 minus the false-negative rate). These definitions are appropriate for studies in which cutoff values of single parameters are used in an attempt to discriminate between groups. Given that this was a clinical study in which masked observers were asked to make assessments of pooled printouts using no set criteria, we defined our rates with regard to the entire pool, and these definitions appear in the article. In other words, we defined false positives in terms of how many times a printout was incorrectly identified as abnormal and false-negatives as how many times a printout from an abnormal eye was called normal, within the context of the entire pool. Table 5 does use the more conventional definitions, with elimination of those suspected to have glaucoma to reduce diagnostic confusion. The demonstrated specificity was 79% with a sensitivity of 92%.

NEIL T. CHOPLIN, MD
DIANE C. LUNDY, MD
San Diego, California

Deep Sclerectomy in Refractory Congenital Glaucoma

Dear Editor:
In their retrospective series, Lüke et al[1] reported that deep sclerectomy was an ineffective procedure in refractory congenital glaucoma. Yet further examination of their data reveals that although they attempted "deep sclerectomy" in 10 eyes, in actual fact, they failed to do so in all but two patients. Out of 10 cases, 4 required conversion to trabeculectomy—one of whom had vitreous loss after iridectomy, 2 of whom required supplemental trabeculotomies, and 2 of whom had macroperforations during surgery. Of the remaining two eyes, "viscocanalostomy" as opposed to deep sclerectomy was performed, and these both developed hyphemas, which are supposedly uncommon with this procedure.[2] Thus, we can only examine these two eyes that apparently had "nonpenetrating" surgery to determine the effectiveness of such in this patient group. Surely this is too small a number to adequately answer the question of efficacy.

It appears that because "identification of Schlemm's canal was not feasible" and "percolation through the trabeculodescemetic membrane was too weak," nonpenetrating surgery was felt to be "not applicable" and required purposeful conversion to a penetrating technique in 6 of 10 cases. Yet, in another study of a similar technique in 12 eyes,[3] this procedure seemed applicable. In that study, Tixier et al found no intraoperative or immediately postoperative complications, and found the success rate to be equivalent to trabeculectomy, with diminished postoperative risks.

In congenital glaucoma, particularly in eyes with previous surgery, Schlemm's canal may be found as far as 5 mm posterior to the limbus and may be fairly scarred—further adding to the significant challenges in approaching these cases. In our experience, we have found it critical to initiate the deep scleral dissection more posteriorly than usual.

The authors state that the intraocular pressure–lowering

effect of deep sclerectomy is independent of external filtration (subconjunctival bleb), rarely occurring in only 5% of patients, and cite an article on viscocanalostomy as a reference.[2] Again, we feel it necessary to distinguish viscocanalostomy from deep sclerectomy with or without an implant. Although both are nonpenetrating surgeries, bleb formation after viscocanalostomy is rare due to tight closure of the superficial flap[2] (which was not performed by the authors in their viscocanalostomies in this report), whereas low, diffuse blebs are commonly found in deep sclerectomy, where the superficial scleral sutures are loosely tied.[4]

The authors should be commended for attempting a technically demanding procedure in these very challenging cases. However, it must be stated that this study was not able to appropriately measure the effectiveness of deep sclerectomy in refractory congenital glaucoma, the stated objective of the article. This study can only conclude that performing deep sclerectomy is technically difficult in these cases. It may be a fact that the authors had the intention of doing deep sclerectomy, but as the saying goes, some roads are paved with good intentions.

We do feel that the role of nonpenetrating glaucoma surgery in congenital glaucoma is yet undefined. In our experience, we have found reasonable success with an improved safety profile using this approach, and we will be publishing our results soon.

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Iqbal Ike K. Ahmed, MD, FRCS(C)
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References

1. Lüke C, Dietlein T, Jacobi PC, et al. Risk profile of deep sclerectomy for treatment of refractory congenital glaucomas. Ophthalmology 2002;109:1066–71.
2. Stegmann R, Pienaar A, Miller D. Viscocanalostomy for open-angle glaucoma in black African patients. J Cataract Refract Surg 1999;25:316–22.
3. Tixier J, Dureau P, Becquet F, Dufier JL. Deep sclerectomy in congenital glaucoma: preliminary results [in French]. J Fr Ophthalmol 1999;22:545–8.
4. Shaarawy T, Karlen M, Schnyder C, et al. Five-year results of deep sclerectomy with collagen implant. J Cataract Refract Surg 2001;27:1770–8.

Author reply

Dear Editor:
In their comment on our retrospective case series, Drs. Shaarawy and Ahmed state that our results contradict those of Tixier et al,[1] who published reasonable success rates after deep sclerectomy in patients with congenital glaucoma. In this context they may have overlooked that in our series all 10 eyes had a history of antiglaucomatous surgery with an average of 2.5 glaucoma interventions (Table 1), as indicated by the term "refractory congenital glaucoma." In the cited series of Tixier et al, most of the eyes had no previous surgery, with only 3 of 12 eyes having two interventions before deep sclerectomy. Thus, this comparison remains at least questionable.

Drs. Shaarawy and Ahmed recommended carefully distinguishing viscocanalostomy from deep sclerectomy due to different bleb rates. They indicated that bleb formation after viscocanalostomy is rare,[2] whereas it is commonly found after deep sclerectomy.[3] However, this substantially contradicts the recent review article[4] from their own study group stating that "50% of patients who underwent a viscocanalostomy had a subconjunctival filtering bleb." We feel that a watertight closure of a thin scleral flap in a buphthalmic eye is more likely a theoretical aim than practically feasible.

Deep sclerectomy is the essential surgical step of nonpenetrating techniques including viscocanalostomy. The first goal of these techniques is to unroof Schlemm's canal by the dissection of the deep scleral flap and thereby achieve a significant aqueous flow into the intrascleral space. In our series of refractory congenital glaucomas, in 60% of eyes this goal was not reached. However, a considerable rate of complications was observed.

Based on these results, we feel justified in drawing attention to the possible risks of nonpenetrating surgery in these rare cases of refractory congenital glaucoma with extremely thin sclera and pathological topography of Schlemm's canal.

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References

1. Tixier J, Dureau P, Becquet F, Dufier JL. Deep sclerectomy in congenital glaucoma: preliminary results [in French]. J Fr Ophthalmol 1999;22:545–8.
2. Stegmann R, Pienaar A, Miller D. Viscocanalostomy for open-angle glaucoma in black African patients. J Cataract Refract Surg 1999;25:316–22.
3. Shaarawy T, Karlen M, Schnyder C, et al. Five-year results of deep sclerectomy with collagen implant. J Cataract Refract Surg 2001;27:1770–1778.
4. Mermoud A. Sinusotomy and deep sclerectomy. Eye 2000; 14(pt 38):531–5.

Dear Editor:
We have a comment on the article on the risks profile of deep sclerectomy for treatment of refractory congenital glaucomas by Luke et al.

The pressure-lowering effect of deep sclerectomy is independent of external filtration only when the superficial scleral flap is tightly secured, as described in Stegmann's viscocanalostomy. External filtration is also absent when the scarring process is so important that the pressure-lowering effect of the surgery is minimal. In the article by Luke et al, only one of the eyes that underwent deep sclerectomy alone, with or without perforations, developed a filtering bleb. It was, moreover, the only one that achieved intraocular pressure (IOP) control. This is the reason why, to optimize all the possible outflow mechanisms past the trabeculodescemetic membrane, we tie our superficial flap very loosely, provided no visible perforation is present.

Of the 10 eyes in this series, only 2 had uneventful pure